Traffic Flow and Speed Assignment

SMRTTECH 4SC3
Nicolas Jao
400450151

1. $v_f = 88$ km/h
   $k_j = 190$ veh/km
   $k_c = \frac{k_j}{2} = \frac{190}{2} = 95$ veh/km
   $v_c = v_f\left(1 - \frac{k_c}{k_j}\right)$
   $v_c = (88)\left(1 - \frac{95}{190}\right) = 44.00$ km/h

2. $TMS = \frac{\sum(n_i v_i)}{\sum n_i}$
   $= \frac{(5)(43) + (4)(70)}{5+4}$
   $= 55.00$ km/h

3. $v = v_f\left(1 - \frac{k}{k_j}\right)$
   $q = kv$
   earlier state: $k = 60$ veh/km
   $v = (72)\left(1 - \frac{60}{180}\right)$
   $= 48$ km/h
   $q = (60)(48) = 2880$ veh/h
   later state: $v = 24$ km/h
   $24 = (72)\left(1 - \frac{k}{180}\right)$
   $k = 180\left(1 - \frac{24}{72}\right)$
   $= 120$ veh/km
   $q = (120)(24) = 2880$ veh/h
   it is the same

4. $q = kv$
   $v = \frac{14.04\,m}{1\,s} \times \frac{1\,km}{1000\,m} \times \frac{60\,s}{1\,min} \times \frac{60\,min}{1\,h} = 50.544$ km/h
   $q = (45)(50.544) = 2274.48$ veh/h
   $|2274.48 - 2268| = 6.48$ veh/h

5. $60 - 90 = -30$ veh/km below critical density

6. $q = kv$
   $v = \frac{11.0\,m}{1\,s} \times \frac{1\,km}{1000\,m} \times \frac{60\,s}{1\,min} \times \frac{60\,min}{1\,h} = 39.6$ km/h
   $q = (65)(39.6) = 2574$ veh/h

7. Greenshields model is a parabola. Flow can be reached at both ends of it: low density, high speed OR high density, low speed.

8. $TMS = \sum \frac{v_i}{n_i}$
   $= \frac{72 + 55 + 49 + 53 + 61 + 60 + 56 + 66}{8} = 59.00$ km/h

9. $SMS = \frac{58 + 57 + 59 + 61 + 56 + 46 + 64}{8} = 55.25$ km/h

10. $q = \frac{590\text{ veh}}{15\,min} \times \frac{60\,min}{1\,h} = 2360$ veh/h

11. $v = v_f\left(1 - \frac{k}{k_j}\right)$
    $k = k_j\left(1 - \frac{v}{v_f}\right)$
    $q = kv$
    $k = (180)\left(1 - \frac{30}{72}\right) = 105$ veh/km
    $q = (105)(30) = 3150.00$ veh/h

12. $v = \frac{14.097\,m}{1\,s} \times \frac{1\,km}{1000\,m} \times \frac{60\,s}{1\,min} \times \frac{60\,min}{1\,h} = 50.7492$ km/h
    $q = (45)(50.7492) = 2283.714$ veh/h
    peak 15 min rate: $|2283.714 - 2268| = 15.714$ veh/h ← less difference, so it's closer to peak 15 min rate
    full hour volume: $|2283.714 - 1940| = 343.714$ veh/h

13. Space mean speed is based on vehicles occupying a road section at one instant. Instant: 8:AM. Road section: 500 m.

14. $k = k_j$
    $v = v_f\left(1 - \frac{k}{k_j}\right)$
    $v = v_f\left(1 - \frac{\frac{k_j}{2}}{k_j}\right)$
    $v = v_f\left(1 - \frac{1}{2}\right)$
    $v = v_f - \frac{v_f}{2}$
    $v = \frac{v_f}{2}$
    $q_{max} = vk$
    $= \left(\frac{v_f}{2}\right)\left(\frac{k_j}{2}\right)$
    $= \frac{v_f k_j}{4}$
    $= \frac{(84)(1320)}{4} = 4620.00$ veh/h

15. $v = \frac{12\,m}{1\,s} \times \frac{1\,km}{1000\,m} \times \frac{60\,s}{1\,min} \times \frac{60\,min}{1\,h} = 43.20$ km/h

16. $TMS - SMS = 70 - 52 = 18.00$ km/h

17. $q = kv$
    $v = \frac{14.14\,m}{1\,s} \times \frac{1\,km}{1000\,m} \times \frac{60\,s}{1\,min} \times \frac{60\,min}{1\,h} = 50.904$ km/h
    $q = (44)(50.904) = 2239.78$ veh/h

18. $q = kv$
    $k = \frac{q}{v}$
    $N = kL$
    $v = \frac{14.0\,m}{1\,s} \times \frac{1\,km}{1000\,m} \times \frac{60\,s}{1\,min} \times \frac{60\,min}{1\,h} = 50.4$ km/h
    $k = \frac{1840}{50.4} = 36.5079$ veh/km
    $N = (36.5079)(0.5) = 18.25$ veh

19. $v = v_f\left(1 - \frac{k}{k_j}\right)$
    $k = k_j\left(1 - \frac{v}{v_f}\right)$
    $k = (180)\left(1 - \frac{20}{72}\right) = 130.00$ veh/km

20. $420 + 490 + 480 + 595 = 1935$ veh

21. highest is 605 veh
    peak = 605 veh

22. $v = v_f\left(1 - \frac{k}{k_j}\right)$
    $v = (72)\left(1 - \frac{65}{180}\right) = 46.00$ km/h

23. $60 < 90 \rightarrow$ uncongested. observed density is below critical density.

24. $q = kv$
    $k = \frac{q}{v}$
    $v = \frac{13.5\,m}{1\,s} \times \frac{1\,km}{1000\,m} \times \frac{60\,s}{1\,min} \times \frac{60\,min}{1\,h} = 48.6$ km/h
    $k = \frac{1860}{48.6} = 38.27$ veh/km

25. $AADT = \frac{N}{365} = \frac{5584500}{365} = 15300$ veh/day

26. % excess $= \frac{TMS - SMS}{SMS} \times 100\%$
    $= \frac{58 - 48}{40} \times 100\%$
    $= 45.00\%$

27. TMS = arithmetic mean of speeds at fixed point
SMS = harmonic mean
TMS ≥ SMS. equal only when all vehicles travel at same speed. TMS counts all vehicle once. SMS gives weight to slower vehicles because they spend more time on road, increasing chances to appear in the snapshot.

28. $v = v_f(1 - \frac{k}{k_j})$
$q = kv$
$v = 70(1 - \frac{20}{100}) = 44$ km/h
$q = (70)(44) = 3080.00$ veh/h

29. TMS because it's recording speeds at a point over a fixed period of time (10 mins).

30. $PHF = \frac{\text{hourly volume}}{4 \cdot \text{peak 15 min count}}$
$= \frac{439 + 490 + 429 + 570}{(4)(570)} = 0.842$

31. (v) $= v_f(1 - \frac{k}{k_j})$
power of both : 1 (linear)
when k ↑, v ↓ : decreasing

32. $k_c = \frac{k_j}{2} = \frac{190}{2} = 95.00$ veh/km

33. $q = kv$
$k = \frac{q}{v}$
$k = \frac{1880}{30} = 62.67$ veh/km

34. $k = \frac{N}{L} = \frac{30}{0.9} = 33.333$ veh/km
$k_{lane} = \frac{33.333}{2} = 16.67$ veh/km/lane

35. $q = kv$
$v = \frac{11.0\,m}{1\,s} \times \frac{1\,km}{1000\,m} \times \frac{60\,s}{1\,min} \times \frac{60\,min}{1\,h} = 39.6$ km/h
$k = \frac{1980}{39.6} = 50$ veh/km
$k_{lane} = \frac{50}{2} = 25.00$ veh/km/lane

36. 9-day ADT $= \frac{151380}{9} = 16820$ veh/day
AADT $= \frac{5369500}{365} = 14700$ veh/day
% above AADT $= \frac{ADT - AADT}{AADT} \times 100\%$
$= \frac{16820 - 14700}{14700} \times 100\% = 14.42\%$

37. $k = \frac{N}{L} = \frac{50}{1.6} = 31.25$ veh/km
$k_{lane} = \frac{31.25}{2} = 15.63$ veh/km/lane

38. $k = k_j(1 - \frac{v}{v_f})$
$= (180)(1 - \frac{24}{72}) = 120$ veh/km
120 > 90 → congested unstable flow

39. AADT = "annual" average daily traffic. The other options do not last a year

40. (q) $= v_f(k - \frac{k^2}{k_j})$
power of density : 2 (parabola)
leading coefficient : −1 (negative → opens downward)

41. $k = \frac{N}{L} = \frac{58}{1.5} = 38.67$ veh/km

42. $k = \frac{N}{L} = \frac{66}{0.8} = 82.50$ veh/km

43. $v = v_f(1 - \frac{k}{k_j})$
$q = kv$
$v = (72)(1 - \frac{50}{180}) = 52$ km/h
$q = (50)(52) = 2600$ veh/h
% of capacity $= \frac{2600}{3240} \times 100\% = 80.25\%$

44. ADT $= \frac{143820}{9} = 15980$ veh/day

45. spot speed. It catches the speed (74 km/h) at a single spot (single location and instant). An instantaneous speed.

46. $v = \frac{14.19\,m}{1\,s} \times \frac{1\,km}{1000\,m} \times \frac{60\,s}{1\,min} \times \frac{60\,min}{1\,h} = 50.94$ km/h
$q = (46)(50.94) = 2343.24$ veh/h
% difference $= \frac{|2343.24 - 2268|}{2268} \times 100\% = 3.32\%$

47. $v_f = \frac{5.11\,m}{1\,s} \times \frac{1\,km}{1000\,m} \times \frac{60\,s}{1\,min} \times \frac{60\,min}{1\,h} = 18.40$ km/h

48. $v_c = \frac{2.59\,m}{1\,s} \times \frac{1\,km}{1000\,m} \times \frac{60\,s}{1\,min} \times \frac{60\,min}{1\,h} = 9.18$ km/h

49. $k_c = \frac{k_j}{2} = \frac{389.2}{2} = 194.6$ veh/km

50. $v_f = \frac{5.11\,m}{1\,s} \times \frac{1\,km}{1000\,m} \times \frac{60\,s}{1\,min} \times \frac{60\,min}{1\,h} = 18.396$ km/h
$k = k_j$
$v = v_f(1 - \frac{k}{k_j})$
$v = v_f(1 - \frac{\frac{k_j}{2}}{k_j})$
$v = v_f(1 - \frac{1}{2})$
$v = v_f - \frac{v_f}{2}$
$v = \frac{v_f}{2}$
$q_{max} = vk$
$= (\frac{v_f}{2})(\frac{k_j}{2})$
$= \frac{v_f k_j}{4}$
$= \frac{(18.396)(389.2)}{4} = 1790$ veh/h

51. $v = \frac{2.26\,m}{1\,s} \times \frac{1\,km}{1000\,m} \times \frac{60\,s}{1\,min} \times \frac{60\,min}{1\,h} = 8.136$ km/h
$q = kv$
$q = (143.88)(8.136) = 1170.61$ veh/h

52. $v = \frac{1.07\,m}{1\,s} \times \frac{1\,km}{1000\,m} \times \frac{60\,s}{1\,min} \times \frac{60\,min}{1\,h} = 3.852$ km/h
$q = kv$
$q = (267.85)(3.852) = 1031.76$ veh/h

53. 143.88 < 194.6 → below critical density

54. 267.85 > 194.6 → above critical density

55. observation A : $|1790 - 1170.61| = 619.39$ veh/h ← less difference, so it's closer to theoretical capacity
observation B : $|1790 - 1031.76| = 758.24$ veh/h

56. see this PDF